CSCI 4200: Operating System Engineering

Assignment 3 — Memory Management: brk(), malloc() an free()

Due Date: April 26, 2026, 23:55

Overview:

In this assignment, you will complete the programs provided in Assign03-Prog.zip to implement:

- Heap system calls (***brk()*** and ***sys_brk()***)
- Dynamic memory management library functions (***malloc()*** and ***free()***)

Part I (40 marks)

You are required to implement the heap system call for brk() at the user level and sys_brk() in kernel mode.

- brk()
    - Complete the code in include/stdio.h to add the marco related to brk()
    - When brk() is invoked, the corresponding system call should be triggered using 13:

      #define _NR_brk 13 /* As defined in stdio.h*/

- sys_brk()
    - Complete the entry in sys_call_table[NR_SYS_CALL] within kernel/global.c so that system call number 13 correctly invokes sys_brk().
    - Implement the logic in kernel/page.c to adjust the heap break. The adjustment should only succeed if the requested address lies within the valid range (between heap_start and heap_limit).

Part II (60 marks)

You are required to implement malloc() and free() in **lib/libc.c** for dynamic memory management.

- Heap Memory Management
    - Maintains a singly linked list of memory blocks within the heap.
    - Each block has a header containing the size, next pointer, and used flag.
- malloc()
    - Search the list for a sufficiently large free block.
    - If no suitable block exists, call brk() to expand the heap.
        - For simplicity, assume the requested size is always **less than 4 KB**, so only one page-level expansion is required.
    - Support block splitting when the free block is larger than the requested size.

- free()
    - Find the corresponding block based on the pointer (ptr) and mark it as free. Merge the released block with its previous and next blocks whenever possible to reduce fragmentation.

Submission Instructions

Submit a single **ZIP file** containing all required source files:

- stdio.h
- global.c
- page.c
- libc.c